

Asher Siddiqui

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EXPERIENCE

Arclab Solutions

Junior Software Engineer | Go, PostgreSQL, Redis, Kafka, Docker Compose, SpiceDB

Remote

Sep. 2025 – Present

- Engineered a service implementing the cgminer API, allowing simulation of arbitrarily large ASIC miner fleets, and decoupling E2E testing from production data.
- Architected an internal resource scoped authorization layer using SpiceDB, enabling secure multi-tenant access control across core platform APIs.
- Engineered a secure automation token system with resource scoping and permission checks, preventing privilege escalation by integrations, and enabling external programmatic access.
- Strengthened internal API security by rolling out mTLS service-to-service protection and a TLS-capable HTTP client, closing trust gaps between services.
- Secured event ingestion by adding Kafka SASL/SCRAM support and updating Docker Compose to provision secure broker users, improving production parity.
- Deployed a Redis-backed rate limiter to enforce abuse protection consistently across instances.
- Directed database evolution by authoring Postgres/Timescale migrations and performance indexes across multiple domains, improving data consistency and query performance.

TAFSOL Technologies

Software Engineering Intern | Go, MySQL, Docker Compose, AWS S3

Karachi, Pakistan (Hybrid)

Mar. 2025 – Aug. 2025

- Optimized authentication response through concurrent processing and query merging, resulting in a reduced response time from $\sim 7s$ to $\sim 1.5s$ and improved auth latency.
- Built a chat system using WebSockets, and reduced delivery time from $\sim 1.3s$ to $\sim 0.4s$ via LRU caching, resulting in reduced message latency.
- Designed and implemented a custom RBAC middleware with per-route permission enforcement, enabling flexible fine grained authorization across APIs.
- Developed quick pay feature using QR codes for internal network payments, reducing steps for transfers by 40%.

Open Source Contributions

Typescript, Node.js, PostgreSQL

- Refactored GitHub OAuth architecture to a database backed source of truth, eliminating fragmented token state, resulting in a cleaner data flow, and more readable codebase. ([PR #1](#), [PR #2](#))
- Designed a provider agnostic context system across Claude, Gemini, Codex, and OpenCode, reconstructing session state on provider switches and branch forks. ([PR #1](#), [PR #2](#), [PR #3](#))
- Built an agentic skills lifecycle system, handling consistency across sandboxes, and global prompt injection to ensure skill awareness across all providers. ([PR #1](#), [PR #2](#))

PROJECTS

FYP - DDoS Detection in Kubernetes using ML and DL | Kubernetes, Ansible, Prometheus, Falco, Go, Python | [GitHub](#)

- Deployed a bare metal Kubernetes cluster and a botnet using Ansible to create simulated DDoS attack scenarios for training ML and DL models.
- Developed a custom data collection pipeline with Falco and Node Exporter to collect system calls and resource metrics, ensuring centralized logging and data generation.
- Developed and deployed attack scripts to simulate and orchestrate attacks via a C2 server and botnet, enabling automated DDoS attacks.

TECHNICAL SKILLS

Languages: Go, Python, Node (Typescript)

Frameworks/Libraries: Echo, Next.js, WebSockets, REST APIs

Databases: MySQL, PostgreSQL, Redis, SpiceDB

Infrastructure: Docker, Docker Compose, Kubernetes, AWS (EC2, S3), Linux, Kafka

EDUCATION

FAST - National University of Computer and Emerging Sciences (NUCES)

Bachelor of Science in Software Engineering

Karachi, Pakistan

August 2021 – June 2025

Awards & Honors: Dean's List of Honors (3.5+ GPA) in Fall 2023, Spring 2024, Fall 2024